



Cloud Computing & DevOps

Certification

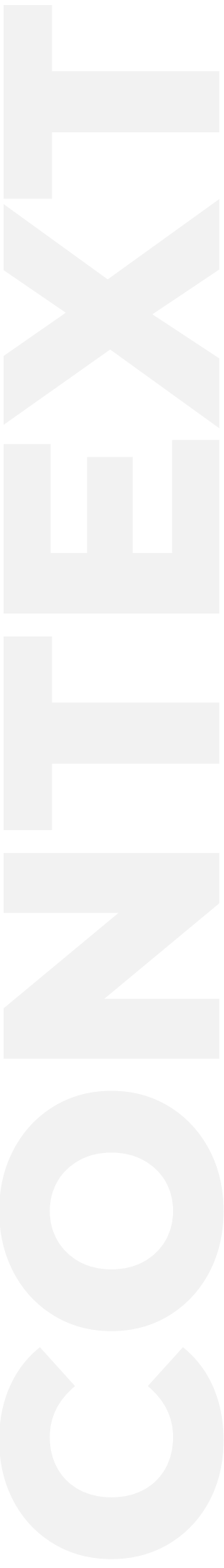
Career Acceleration
Program

In collaboration with



www.learnbay.co



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Advance Cloud Computing and Devops

Embark on a transformative journey with LearnBay's comprehensive Cloud Computing & DevOps Program.

This course is crafted to empower you with expertise in both Cloud Computing and DevOps methodologies. Dive into a rich learning experience with extensive hands-on training in key platforms like **AWS, Azure, GCP and various DevOps tools.**



35k+

Trusted
Learners



2cr

scholarship
s awarded



600+

professionals
secured jobs

Program Highlights



Industry-Relevant & Updated Syllabus

Learn the industry's latest tools, techniques & trends. Gain hands-on experience developing various apps.



360 Degree Knowledge Building

Develop practical skills through real-world projects and assignments



1:1 Dedicated Mentorship

Personalized learning experience from experienced industry professionals.



Multiple Career Opportunities

We don't just educate; we guide you towards your career goals with our dedicated placement pool.

Talk to our counsellor

- Curriculum Insights
- Post-Program Support
- Financing Solutions
- Career Prospects Exploration

[**Inquire Now**](#)



Program Details

COURSE PREREQUISITE

Basic understanding of IT and software with foundational knowledge in networking is required

PROGRAM ELIGIBILITY

- ✔ **IT Professionals:** Ideal for IT administrators, system admins, network engineers looking to upgrade their skills.
 - ✔ **Software Developers:** Beneficial for developers aiming to specialize in DevOps practices and cloud solutions.
 - ✔ **Career Changers:** Open to professionals from different backgrounds seeking to transition into the IT and cloud domain.
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KEY FEATURES

- ✔ Practical exercises, real-world projects, and lab sessions using tools like AWS, Azure, Kubernetes, Docker, and Jenkins.
 - ✔ Dedicated Placement Cell | 100% Guaranteed Interview calls
 - ✔ Globally Recognised **Certification from IBM & Microsoft**
-

JOB ROLES TO TARGET

Cloud Engineer
DevOps Engineer
Site Reliability Engineer (SRE)
System Administrator

Cloud Architect
Automation Engineer
Infrastructure Engineer
Cloud Security Specialist

[Click below](#)

[Check Eligibility](#)

Learnbay's ProjectLab

Choose Learnbay for your career journey because we're more than just a training provider. Our Project Innovation Lab lets you apply your skills in real-world scenarios. Get **dual certifications** for a competitive edge. Discover how Learnbay can boost your career growth. Don't settle for less – choose Learnbay, your path to success!

1 Project Innovation Lab

Work in an industry like environment and gain practical hands-on experience of cloud & devops with dedicated mentors from industry.

2 Dedicated Placement Cell

Experience 100% job assistance with guaranteed interview call from leading MNCs and startups globally.

3 Dual Certification

Gain top-notch skills for a successful career through our certification program in collaboration with Microsoft

1 Project Innovation Lab

Learnbay's Project Innovation Lab replicates industry like environment for real time projects. With our **ProjectLab**, you gain real proof of hands-on experience by having your project certified by the industry.

In our ProjectLab, you work like a cloud engineer with dedicated project mentors from industry and get certified on capstone project.

450+

Hiring
Partners

 PUNE



35k
Trusted
Learners



1-1 Doubt Session

 HYDERABAD



**Capstone
Project
Certificate
from IBM**



 DELHI



 BENGALURU

**Project
Innovation Labs
Across India**

2 Career Service PRO



Get 3 years of Job and Placement support

Unleash your career potential with unlimited job access, interview support, and profile review.

6 Mock Interviews with Industry Experts

Master the art of cloud computing, devops and stay ahead of the curve with mockups and industry insights



Resume Building Session

Craft a powerful resume showcasing your expertise in software development to stand out from the competition

1:1 LinkedIn Review Session

Get 1-on-1 sessions with experts for a better understanding of LinkedIn profile reviews and cover letter.



Dedicated Placement Cell

Receive unlimited interview calls from a diverse pool of interested employers/recruiters until you successfully secure a job

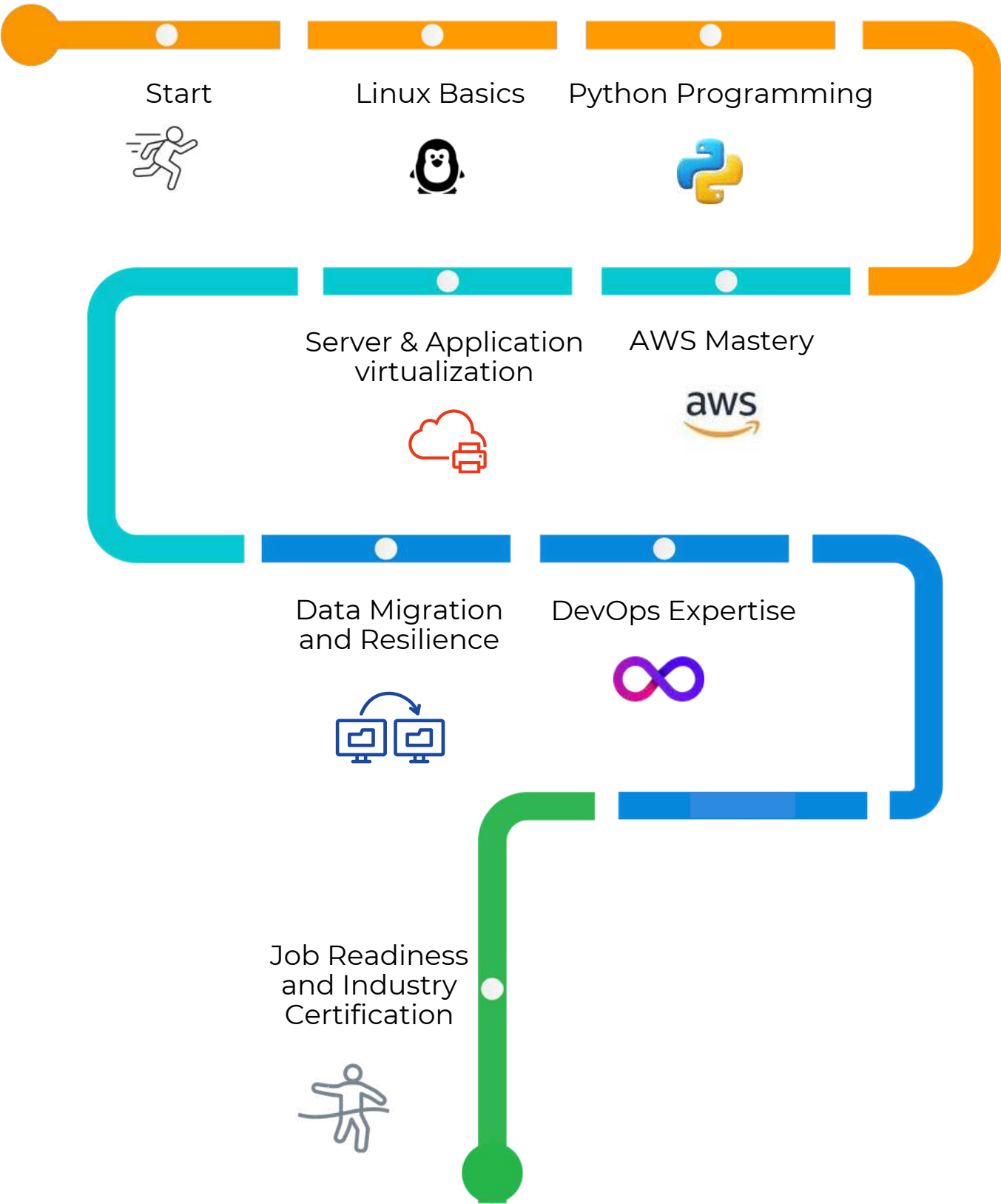
3 Industry Certification

Microsoft Azure Certificate

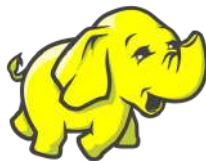
Achieve professional growth & increase earning potential with Microsoft Azure certification



Learning Path



Tools Covered



Others Vs Learnbay

Benefits		Learnbay	Others
	Guaranteed Interview Calls	☒	☐
	Industry capstone project certificate from IBM	☒	☐
	Domain specialized programs for professionals	☒	☐
	100% live interactive sessions with industry experts	☒	☐
	On-demand video call with industry experts	☒	☐
	Personalised Resume Review Session	☒	☐

Program Fee & Financing



Scholarship



Scholarships are awarded based on profile review. Eligible candidates can avail upto 25% scholarship on desired courses. Click the button below to apply.

[Click below](#)

[Check Scholarship Eligibility](#)

Financing as low as
Rs. 5,244/month

No Cost EMI

 **Razorpay**


LIQUILOANS

 **Propelld**

 **ShopSe**

Program Fee

Rs. 80,000/- +18% GST

Linux Fundamentals and Introduction 10 Hours

Linux Basics and System Administration

- Introduction to Linux Operating System
- Command Line Interface (CLI) Essentials
- User and Group Management
- Networking Basics in Linux

Practical exercises and assignments for mastering Linux commands

- Command Line Mastery: Exercises focusing on everyday CLI usage
- Script Writing: Basics of Shell Scripting for Automation
- System Administration Tasks: Setting up a Web Server, Creating Cron Jobs

Tools covered



LINUX



Python Programming Essentials

25 Hours

Introduction to Python and Environment Setup

- Basics of Python Programming Language
- Setting Up Python Development Environment
- IDE Setup and Introduction (PyCharm, Jupyter Notebook)

Python Programming Foundations

- Data Types, Variables, and Basic Operators
- Control Structures: If-Else Statements, Loops
- Functions: Definition, Arguments, Return Values

Tools covered



jupyter



PyCharm

Data Handling and Structures

- Lists, Tuples, Dictionaries, and Sets
- String Manipulation and Regular Expressions
- File Operations: Reading and Writing Files

Real time project will be covered after Python completion with 2 hours of assessment (coding and objective questions)



Python Programming Essentials

Advanced Python Concepts

- Object-Oriented Programming (OOP) in Python
- Classes, Objects, Inheritance, and Polymorphism
- Encapsulation and Abstraction
- Exception Handling and Debugging Techniques
- Modules and Packages: Importing and Creating

Project Work and Practice

- Hands-on sessions with Python scripting and automation projects
- Real-world problem-solving using Python
- Mini-projects: Developing small-scale applications or scripts

AWS Cloud Mastery

40 Hours

AWS Administration Level

Introduction to AWS

- AWS Services Overview
- Setting Up AWS Accounts
- AWS Management Console

AWS Storage Services

- S3 (Simple Storage Service)
- EBS (Elastic Block Store)
- Glacier
- Storage Gateway
- Amazon FSx (Fully Managed File Storage)

Networking and Content Delivery

- VPC (Virtual Private Cloud)
- CloudFront
- Route 53
- API Gateway
- AWS Direct Connect

Monitoring and Management

- CloudWatch
- CloudTrail
- AWS Config
- AWS Trusted Advisor

AWS Compute Services

- EC2 (Elastic Compute Cloud)
- AWS Lambda
- Elastic Beanstalk
- Lightsail
- Amazon ECS
- EKS (Elastic Kubernetes Service)

AWS Database Services

- RDS (Relational Database Service)
- DynamoDB
- Redshift
- ElastiCache
- Amazon Aurora

Automation and Optimization

- AWS CLI (Command Line Interface)
- CloudFormation
- Systems Manager
- AWS Elastic Beanstalk for Deployment Automation

AWS Cloud Mastery

Transitioning to Solution Architect

Architecting on AWS

- Designing Systems (Availability, Cost-Efficiency, Fault-Tolerance, Scalability)
- Decoupling Mechanisms and Scalable Design
- Choosing Appropriate AWS Services

Advanced Networking

- VPC Design and Implementation
- Advanced Security and Networking Best Practices

High Availability and Business Continuity

- Disaster Recovery Strategies
- High Availability Approaches

Migration and Innovation

- Migration Strategies to AWS
- Leveraging AWS for Innovation

Data Security

- Data Encryption Methods
- Compliance and Governance

Cost Optimization

- Cost-Effective Resource Utilization
- Monitoring and Optimization Techniques

Designing for New Technologies

- IoT (Internet of Things) Solutions
- Machine Learning Implementations
- Serverless Architectures

AWS Cloud Mastery

AWS Certified Solutions Architect - Associate

Exam Preparation

Preparation Tips

- Example: "AWS Whitepapers and FAQs Review" - A comprehensive study of AWS official documentation, focusing on key services like EC2, S3, RDS, and VPC.
- Case Study: "Migration of Legacy Systems to AWS" - This case study discusses the process and considerations for migrating traditional data centers to AWS services.

Practice Exams and Scenarios

- Example: "Scalable Web Application Architecture on AWS" - A scenario that requires designing a highly scalable and reliable architecture for a web application using AWS services.
- Case Study: "Disaster Recovery Setup for E-Commerce Platform" - A scenario focused on building a disaster recovery plan for an e-commerce platform using AWS services like Route 53 and S3.

AWS Certified Solutions Architect - Professional

Deep Dive into Complex Architectures

Detailed Study of Advanced AWS Topics

- Example: "Implementing Serverless Architecture" - An in-depth exploration of serverless computing using AWS Lambda and API Gateway.
- Case Study: "Enterprise-Level Data Warehousing with Redshift" - A detailed case study on setting up and optimizing a data warehouse for a large enterprise using Amazon Redshift.

Microsoft Azure Proficiency

50 Hours

Azure Administration Level

Introduction to Azure

- Overview of Azure Services
- Setting Up Azure Accounts
- Azure Portal and Azure CLI

Azure Storage Solutions

- Blob Storage
- Azure Files
- Queue Storage
- Disk Storage
- Azure Data Lake Storage for Big Data Analytics

Identity and Access Management

- Azure Active Directory
- Role-Based Access Control (RBAC)
- Multi-Factor Authentication
- Azure AD Connect

Security and Compliance

- Azure Security Center
- Key Vault
- Azure Policy
- Azure Blueprints

Azure Compute Services

- Azure Virtual Machines
- Azure Kubernetes Service (AKS)
- Azure Functions
- App Services
- Azure Container Instances for Container Deployments

Azure Networking

- Virtual Networks (VNet)
- Load Balancers
- VPN Gateway
- Azure DNS
- Azure ExpressRoute for Private Connectivity

Database Services

- Azure SQL Database
- Cosmos DB
- Azure Database for MySQL
- Azure Database for PostgreSQL
- Azure Synapse Analytics for Big Data and Analytics

Monitoring and Management

- Azure Monitor
- Azure Log Analytics
- Azure Automation
- Azure Backup and Site Recovery



Microsoft Azure Proficiency

Transitioning to Solution Architect

Architecting Solutions on Azure

- Designing Infrastructure Solutions
- Designing for Security and Identity
- Designing Data Platform Solutions
- Designing Business Continuity

Advanced Networking

- Network Architecture and Strategies
- Azure Application Gateway
- Network Security Groups

Data and Databases

- Advanced Database Management
- Big Data and Analytics Solutions
- Data Migration Strategies

Developing for the Cloud

- Azure DevOps
- CI/CD Pipelines
- Azure App Service Environments

Application Modernization

- Microservices and Containers
- Serverless Computing
- Hybrid Cloud Solutions

Cost Management and Optimization

- Microservices and Containers
- Serverless Computing
- Hybrid Cloud Solutions



Microsoft Azure Proficiency

Preparation for Azure Architect Technologies

- Certification Path and Exam Preparation
- Practice Tests and Scenarios

Advanced Course: Azure Solutions Architect Expert

- Complex Architectural Solutions
- Case Studies and Real-World Scenarios
- Advanced Certification Preparation

Case Studies and Projects

- "Building Scalable Web Applications on Azure": Designing and deploying a scalable web application using Azure App Services and Azure SQL Database.
- "Hybrid Cloud Setup for Enterprise": Implementing a hybrid cloud solution for an enterprise using Azure Stack and Azure Cloud Services.
- "Data Lake and Big Data Analytics on Azure": Setting up a Data Lake using Azure Data Lake Storage and analyzing big data with Azure Synapse Analytics.
- "Modernizing Legacy Applications with Azure": Migrating and modernizing legacy applications to the Azure cloud, leveraging Azure Kubernetes Service and Azure DevOps for continuous integration and delivery.

Google Cloud Platform

50 Hours

GCP Administration Level

Introduction to Google Cloud Platform

- Overview of GCP Services
- Creating and Managing GCP Accounts
- Google Cloud Console

Storage and Database Services

- Cloud Storage
- Cloud SQL
- Cloud Spanner
- Firestore
- Bigtable
- Firebase Realtime Database for Mobile and Web Apps

Identity and Security

- Cloud Identity and Access Management (IAM)
- Security and Compliance
- Key Management Service (KMS)
- Security Command Center
- Cloud Armor for Network Security

API Management

- Apigee API Platform
- Endpoints
- Cloud Run

Compute Services

- Compute Engine
- Google Kubernetes Engine (GKE)
- App Engine
- Cloud Functions
- Cloud Run for Serverless Applications

Networking Services

- Virtual Private Cloud (VPC)
- Cloud Load Balancing
- Cloud CDN (Content Delivery Network)
- Cloud Interconnect and Peering
- Cloud VPN for Secure Connectivity

Monitoring and Operations

- Cloud Monitoring
- Cloud Logging
- Error Reporting
- Cloud Trace and Cloud Debugger

Developer Tools

- Cloud SDK
- Cloud Shell
- Cloud Source Repositories



Google Cloud Platform

Transitioning to Solution Architect

Architecting with Google Cloud Platform

- Designing Infrastructure Solutions
- Designing for Security and Identity
- Designing Data Platform Solutions
- Designing Business Continuity

Advanced Networking

- Hybrid and Multi-cloud Networking
- Network Design and Optimization
- Network Security Best Practices

Data Management and Databases

- Advanced Database Services
- Big Data Solutions
- Migration and Management Strategies
- Data Catalog for Metadata Management

Data Management and Databases

- High Availability and Disaster Recovery
- Auto-scaling and Load Balancing
- Best Practices for Reliability

Security, Compliance, and Identity

- Advanced Security and Compliance
- Identity Federation and Management
- Data Protection and Encryption

Cost Management and Optimization

- Cost Control and Reporting
- Efficient Resource Utilization
- Billing and Budget Management



Google Cloud Platform

Machine Learning and Artificial Intelligence

AI Platform

- Machine Learning Services
- Integrating AI into Cloud Solutions
- AutoML for Custom Model Training

Case Study:

"Retail Demand Forecasting with GCP AI and ML Services" - Utilizing GCP's machine learning capabilities to predict product demand for a retail chain.

Case Studies and Projects

- "Implementing Secure Hybrid Networking on GCP": A project focused on setting up a secure hybrid network using GCP's VPC, Cloud Interconnect, and VPN services.
- "Building a Scalable E-Commerce Platform with GCP": Case study on designing and implementing a highly scalable and reliable e-commerce platform using GCP services.

Server and Application virtualization

16 Hours

Overview of Virtualization

- Introduction to Virtualization: Understanding Its Historical Development
- Different Forms of Virtualization

Server Virtualization

- Introduction to Server-Based Virtualization
- Understanding Hypervisor-Based Virtualization
- Comparing Type 1 and Type 2 Hypervisors
- Differences Between Full and Paravirtualization

Virtual Machines (VMs)

- The Lifecycle of a Virtual Machine
- Structure and Formats of Virtual Machine Images
- Detailed Look at Virtual Machine Instances: Public Cloud Specificities
- Resource Allocation for Virtual Machine Instances
- Networking Within Virtual Machines: An Overview
- Exploring Amazon Machine Instances

Application Virtualization

- Application Virtualization in VMs

Tools covered



Microsoft
Application
Virtualization

citrix

Virtual Apps
and Desktops

Server and Application virtualization

Containerisation

- Fundamentals of Container-Based Virtualization
- Lifecycle of Containers
- Container Image: Structure and Formats
- Managing Container Instances: Resource Allocation
- Networking in the Context of Containers

Unikernels

- Unikernels: Composition, Formats, and Management
- Allocating Resources and Networking in Unikernels

Comparative Analysis

- Differentiating Between VMs, Containers, and Unikernels

Network Function Virtualization (NFV)

- An Introduction to Network Function Virtualization

Tools covered



TURBO.NET



Data Migration and Resilience

12 Hours

Introduction to Data Migration in Cloud Environments

- Understanding the Basics of Data Migration
- Key Considerations for Cloud Data Migration

Data Migration Strategies and Best Practices

- Planning and Executing a Migration Strategy
- Tools and Techniques for Effective Data Migration

Building Resilience and High Availability in the Cloud

- Principles of Resilience in Cloud Computing
- Designing for High Availability and Disaster Recovery

Integrating Data Migration with DevOps Practices

- Automation in Data Migration Processes
- Continuous Integration and Continuous Deployment (CI/CD) for Data Migration

Case Studies and Real-world Applications

- Analyzing Successful Data Migration Projects
- Lessons Learned and Best Practices in Action

Tools covered

IBM DataStage

rsync



rubrik

Zerto



DevOps Expertise

Introduction to DevOps

DevOps Overview

What is DevOps, Benefits of DevOps, DevOps lifecycle, Key principles

DevOps Culture and Collaboration

Differences between DevOps and traditional models, DevOps culture and mindset, Team roles

Version Control

Introduction to Version Control

Importance of version control, Version control systems (VCS)

Git Basics

Git setup, Git workflow, Branching and Merging, Tagging, Git commands

GitHub & GitLab

Repository management, Collaboration, Pull Requests, Webhooks

Continuous Integration (CI)

CI Concepts and Importance

What is CI, Benefits of CI, Overview of CI tools

Jenkins

Jenkins installation, Jenkins Pipeline, Jobs, Plugins, CI/CD pipelines

#

GitLab CI configuration, YAML syntax, Runners, GitLab CI/CD pipeline



DevOps Expertise

Build and Release Management

Build Automation

Introduction to build automation, Build tools (Maven, Gradle), Build automation for CI

Package Management

Package repositories (Nexus, JFrog Artifactory), Managing dependencies

Release Management

Release strategies, Managing releases in Git, Versioning

Continuous Delivery (CD) & Deployment Automation

CD Overview and Tools

Continuous delivery principles, Differences between CI and CD

Deployment Automation

Infrastructure as Code (IaC), Benefits of automation, Tool overview (Ansible, Terraform)

Ansible Basics

Ansible playbooks, Inventory management, Modules, Roles

Terraform

Terraform basics, Writing configuration files, State management



DevOps Expertise

Container Orchestration Basics

Introduction to Containers

What are containers, Benefits of containerization, Introduction to container orchestration

Kubernetes

Kubernetes architecture, Pods, Services, Deployments, Namespaces, ConfigMaps, Secrets

Helm

Helm basics, Charts, Repositories, Managing Helm releases

Continuous Monitoring

Monitoring and Logging Concepts

Importance of monitoring, Metrics, Logging basics

Prometheus and Grafana

Prometheus installation, Metrics collection, Grafana dashboards

ELK Stack (Elasticsearch, Logstash, Kibana)

Setting up ELK, Log collection and processing, Visualizing logs in Kibana

Executive-level real-time **Industrial Projects**

#1

CI/CD Pipeline for a Microservices Application

Create a multistage CI/CD pipeline for deploying a microservicesbased application to a cloud platform like AWS, Azure, or Google Cloud Platform (GCP).

Tools: Jenkins, GitLab CI/CD, Docker, Kubernetes, Helm, Terraform, Ansible.

Outcome: Automated build, test, and deployment using Jenkins or GitLab CI/CD.

#2

Cloud Infrastructure Automation with Terraform

Design and deploy an automated cloud infrastructure setup using Terraform to create and manage cloud resources like virtual machines, databases, and storage services.

Tools: Terraform, AWS CloudFormation, Azure Resource Manager (ARM) templates.

Outcome: Create reusable modules for infrastructure components.

#3

Monitoring and Logging Solution on the Cloud

Set up a robust monitoring and logging infrastructure for realtime data analysis and alerting.

Tools: Prometheus, Grafana, ELK Stack (Elasticsearch, Logstash, Kibana), AWS CloudWatch, Datadog.

Outcome: Collect logs and metrics from applications and infrastructure.

#4

Serverless Application Deployment with DevOps Practices

Build and deploy a serverless application using cloud functions, ensuring that the entire process follows a DevOps workflow.

Tools: AWS Lambda, Azure Functions, Google Cloud Functions, GitHub Actions, Serverless Framework.

Outcome: Deploy serverless functions as part of a CI/CD pipeline.

Executive-level real-time **Industrial Projects**

#5

CloudBased DevOps Pipeline with Security Integration (DevSecOps)

Implement a secure DevOps pipeline that incorporates security practices (DevSecOps) into the development and deployment cycle.

Tools: Jenkins, Azure DevOps, GitLab CI/CD, Docker, OWASP ZAP, SonarQube.

Outcome: Integrate security scanning into the build pipeline.

Thank you!

For more queries and information
please reach out to us at:

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Visit us at

www.learnbay.co

